

Calibrated Autonomy for Experimental Protocol Agents

Research Exposé, ETH Agentic Systems Lab, AI Industry Transformation Track

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The Problem

During my materials research at Carnegie Mellon University, I spent weeks running density functional theory (DFT) simulations with Quantum ESPRESSO on high-performance computing clusters. Every completed simulation produced energies, convergence logs, charge densities, and magnetic configurations. What it never produced was the reasoning behind the next experiment.

Each simulation changed what I believed about the material I was studying. Some results strengthened a hypothesis, others contradicted it, and some introduced entirely new questions. Those evolving beliefs determined which simulation I launched next, yet they remained scattered across notebooks, conversations, and my own memory. The computational workflow recorded outputs. It did not record the scientific reasoning that connected one experiment to the next.

Agentic systems built for experimental design, including protocol agents like the one BIIE AI is building as part of its Scientific Research Assistant product, are starting to close that gap by chaining literature search, experiment planning, and result interpretation into a single automated loop. But most of these agents are still evaluated on task completion: did it finish the protocol, did it return an answer. Very few are evaluated on whether they knew when their own evidence was too weak, too contradictory, or too exhausted to justify the next action.

An agent that can run experiments but does not know when to stop running them has not actually reduced the researcher's judgment burden. It has just moved that judgment downstream, to whoever has to audit the agent's conclusions after the fact.

Why It Matters

In immunology and biomedical research, as in materials science, computational and wet-lab experiments are expensive and the hypothesis space is large. An agent that keeps acting past the point its evidence justifies wastes budget on low-information experiments and, worse, produces conclusions a scientist cannot trust without re-deriving them independently. For a product like BIIE's Scientific Research Assistant, which is meant to support scientists across the full research lifecycle rather than replace their judgment, this is not a peripheral concern. Trustworthy automation of experimental design depends on agents that can recognize the limits of their own current understanding, not only execute a plan competently.

What I Would Build

A protocol agent that maintains an explicit, inspectable basis for every decision it makes throughout a long-running scientific investigation: what it currently believes, how those beliefs have changed, what evidence supports them, and when uncertainty justifies further investigation instead of continued autonomy.

The mechanism is intentionally domain independent. it weighs labeled candidates against a measured value and a noise floor, not materials-science-specific concepts, so the same logic applies wherever competing outcomes must be judged against a margin of confidence. Domain-specific tools generate observations, while the agent maintains a persistent representation of the evolving scientific investigation, updates competing hypotheses as evidence accumulates, records how each observation changes its confidence, and determines whether it should continue, gather additional evidence, or escalate to a human researcher.

Over the course of a larger research program, the agent would accumulate calibration knowledge across many investigations rather than treating each experiment independently. It would recognize recurring failure patterns, reuse previously validated diagnostic strategies, and gradually improve the quality of its own autonomous decision making while preserving a complete reasoning trail that researchers can inspect and override.

Although this proposal is motivated by computational materials science, the same architecture naturally extends to experimental protocol agents such as BIIE's Scientific Research Assistant, where scientific reasoning similarly unfolds through sequential evidence accumulation under uncertainty.

Proof of Concept (this submission)

The proof of concept demonstrates the proposed architecture in computational materials science using data from my existing DFT+U study of binary (Mn,X)WO₄ tungstates to evaluate how an agent maintains and updates its internal reasoning throughout an investigation. It consists of three complementary components:

First, the agent evaluates a convergence-monitoring policy using a real completed simulation trace. The calculation is replayed iteration by iteration, reflecting how evidence would become available during execution, and the agent decides whether the current trend justifies continuing, concluding, or escalating for review. Its decisions are evaluated against the final outcome reported by the simulation.

Second, the agent constructs a persistent calibration record from completed investigations. Instead of treating every failure or ambiguous result as an isolated event, it distinguishes recurring failure patterns from fundamentally inconclusive outcomes, allowing previously acquired diagnostic knowledge to influence future decisions while preserving the evidence supporting each conclusion.

Finally, the agent applies this accumulated calibration knowledge to previously unseen cases. When sufficient precedent exists, it generates a justified recommendation together with the reasoning that supports it. When no defensible analogy can be established, it explicitly represents that uncertainty and recommends acquiring new evidence rather than producing an unsupported conclusion.

To evaluate the contribution of calibrated autonomy, the prototype also compares these decisions against a baseline agent lacking persistent calibration knowledge, enabling evaluation of whether explicit uncertainty modelling produces more trustworthy autonomous behaviour.

For every rerun, recheck, or new-candidate case it decides on, the agent generates a real, submittable Quantum ESPRESSO input file and attempts to run it. This environment has no cluster access, so that attempt fails visibly rather than being hidden, and the input is left ready for a human or job scheduler to submit.

The PoC implements this as a deterministic decision layer: a failure-signature playbook that accumulates across investigations, and threshold-based rules that classify each result as conclusive, worth rechecking, or genuinely inconclusive. Every classification is traceable to the specific rule and evidence that produced it. The thesis would extend this into a persistent, structured belief representation, confidence over competing hypotheses that updates as evidence accumulates, rather than the current flat lookup table, and would replace the fixed thresholds with model-based reasoning over unstructured evidence such as literature.

Research Questions

RQ1: When an agent has no calibration precedent for a new case, which fallback strategies (nearest-analog matching by underlying physical property, explicit uncertainty labeling, escalation to a fresh baseline measurement) produce recommendations a domain expert would actually trust?

RQ2: Does an agent with an explicit escalate-when-uncertain decision layer and a persistent record of previously diagnosed failures produce more trustworthy conclusions than an otherwise identical agent that reports every result as fact and re-diagnoses every failure from scratch?

RQ3: How do the analogy-matching and escalation strategies learned in this materials-science setting transfer to a real domain such as BIIE’s immunology protocols, where the cost of a wrong unflagged prediction is far higher than in a materials screening search?

Fit and Next Steps

This direction extends a pattern across my prior work: Quilltale separates story generation from world-state management so reasoning stays traceable; my evaluation projects distinguish failure types instead of reporting a single accuracy number. BIIE’s Experimental Protocol Agent needs the same instinct applied to real scientific workflows. I would welcome the chance to develop this as an Associated Researcher within the AI Industry Transformation Track, working directly with BIIE to extend it toward their own protocols. I am available to start as soon as possible and able to relocate to Zurich.